

GoalCoach: Proposal & Senior AI Engineering Review

Complete Translation, Unvarnished Architecture Critique & Blueprint for the 4-Week Sprint

Author: Senior AI Engineer & Systems Architect (30 yrs. industry exp.) • **Project:** GoalCoach (HSK Chinese Multi-Agent Learning System) • **Focus:** Pragmatism, 4-Week Timeline, OpenRouter \$50/week & Local Gemma Backup

PART 1: COMPLETE PROJECT PROPOSAL (ENGLISH TRANSLATION)

1. The Proposed Application

GoalCoach is an agentic AI learning coach that helps learners master Chinese exam preparation (with a special focus on HSK) through continuous, personalized planning rather than isolated conversations.

Existing Chinese-learning products fall into two categories:

- **Structured platforms** (e.g., HelloChinese, SuperChinese) offer systematic courses, but fix the learning path and provide little genuine personalization beyond re-serving missed exercises.
- **Conversational AI tutors** (e.g., ChatGPT) are flexible, but lack long-term planning: learners must know what to ask, progress is not tracked across sessions, and learning quickly becomes fragmented.

GoalCoach combines the strengths of both worlds: based on the learner's goal (e.g., passing HSK4, business Chinese), learning history, and test performance, the system continuously generates a dynamic, personalized daily study plan. It proactively guides the learner on what to study today instead of waiting for prompts or imposing a rigid curriculum.

Target Audience: Chinese-as-a-second-language learners, especially HSK exam candidates. The architecture is designed to easily scale later to additional languages, corporate training, and other goal-driven long-term learning tracks.

Key Feature: A transparent overall progress view. Every day, the learner sees the exact percentage of their goal (e.g., HSK4 vocabulary and grammar) they currently master. This is not a permanent, purely increasing counter; it honestly reflects both what has been learned and what is at risk of being forgotten. It is computed via LLM-graded exercises rather than basic right/wrong checks.

2. The Agentic System

GoalCoach is not a static pipeline. The learner's state (goal, review schedule, plan validity, and current session) determines which specialized agent executes next. The system deliberately splits into a deterministic control layer and reasoning agents, rather than routing every single decision through an expensive LLM:

Workflow Orchestrator

Acts as a lightweight, deterministic controller without an LLM. Routing does not require logical inference, saving model calls. On every visit, it inspects the learner's state and delegates by fixed priority:

1. Goal newly set or updated → **Goal Planning Agent**
2. Scheduled concept review due → **Progress & Mastery Agent**
3. Daily plan invalid or completed → **Goal Planning Agent**
4. Default case → **Teaching Agent** (to execute today's study plan)

Goal Planning Agent

Activated only when a goal is set or modified; breaks down the goal into milestones and a roadmap. During the initial onboarding session, it administers a brief diagnostic assessment to initialize mastery levels and milestones.

Progress & Mastery Agent

Updates four learner-specific states following each LLM-graded exercise:

- **Mastery Score:** How proficiently a concept was demonstrated during evaluations.
- **Retention Probability:** Estimated retention chance derived from a spaced-repetition forgetting curve. Decays over time, strengthens on correct answers, and resets on errors.
- **Personal Error Profile:** Recurring grammar and vocabulary mistakes (e.g., confusing 了 vs. 过, or specific measure words).
- **Next Review Schedule:** Due dates per concept governed by the retention model.

Computes overall progress as $\Sigma (Mastery \times Retention)$ across all target concepts. A concept learned a month ago loses weight if left unreviewed.

Retrieval Agent

Conditionally invoked when knowledge gaps arise or new material is required. Autonomously determines if retrieval is necessary or if cached content can be reused, crafts targeted search queries based on the error profile, and re-ranks ChromaDB vector results.

Teaching Agent

Generates open-ended, LLM-graded exercises (no multiple choice), explains concepts, conducts practice dialogues, and adaptively adjusts difficulty mid-session. Generates session summaries and seeds the plan for the next day.

3. The Technical Approach

Orchestration via **LangGraph** with conditional edges for deterministic routing. Tool calling and structured outputs (**Pydantic**) ensure reliable data exchange (goals, plans, mastery scores, error profiles). **ChromaDB** serves as the vector store for curated HSK material. **OpenAI API** powers reasoning, dialogue, and open-ended grading. A **FastAPI** backend connects to a lightweight web frontend.

Grading reliability as a first-class problem: Evaluation is conducted across explicit rubric dimensions (grammar, semantics, register). Validation against 20–30 human-annotated reference examples per HSK level prevents LLM hallucinations. Weeks 1–2 focus on the core branching workflow and state management, with UI and dashboards built in subsequent weeks.

4. Value & Differentiation

Solves the drop-off issue in language apps by removing planning friction through a singular, unambiguous daily study objective. Provides an honest progress metric rather than gamified pseudo-streaks. The modular architecture readily adapts to enterprise training and other languages.

5. The Team

Name	Role	Relevant Skills & Background
Weijia Han	AI / Agent Engineer (Team Lead)	Project management at Bigo & Huawei, data annotation, linguistics & translation, 42 Heilbronn (Python, RAG, Agentic AI).
Jiyan Wang	Backend Engineer	42 Heilbronn (C/C++, HTTP server implementation, Minishell, Docker/Inception, FastAPI, REST API design, databases, system integration).
Weng Man Cheung	AI / Data Engineer	Translation & Lexicography, linguistics, 42 Heilbronn (LLMs, Python, Schwarz IT AI Agent workshop, language corpora, XML/HTML).
Musab	AI Engineer & Core Systems Architect	Automotive engineering & AI specialist with deep Python expertise, Computer Vision, local inference & Edge AI pipeline architectures, multi-agent state orchestration, and structured validation systems (Pydantic / FastAPI).

PART 2: SENIOR AI ENGINEER DEEP-DIVE & ARCHITECTURE REVIEW

Senior Verdict: This concept belongs in the top echelon of academic agent proposals. Avoiding an "LLM-overloaded router system" in favor of a deterministic state machine is an exemplary engineering decision. However, significant execution pitfalls exist for a 4-week sprint.

1. What is Excellent in the Proposal? (Strengths)

BEST PRACTICE Hybrid Routing (Deterministic vs. Reasoning): Nearly 80% of failed agent projects break down because every routing choice is delegated to an LLM. Keeping the top-level orchestrator deterministic (if-else / state machine) saves massive amounts of tokens, cost, and latency.

PRODUCT LOGIC Progress Metric $\Sigma (Mastery \times Retention)$: A genuine differentiator. It breaks away from Duolingo's manipulative "always-up" streak gamification and delivers honest pedagogical value.

SCOPING Curated Knowledge Corpus (20–30 Resources): Recognizing that content volume is secondary to system architecture demonstrates mature engineering discipline.

2. Critical Bottlenecks & Risks (What Must Be Improved)

CRITICAL **Over-Agentification & Latency Explosion:** If the Goal, Progress, Retrieval, and Teaching agents fire sequentially for a single simple exercise, the user will wait 15–25 seconds for a response. *Solution:* Reduce to 1 synchronous execution loop in the frontend; execute background state updates asynchronously!

BUDGET RISK **OpenRouter \$50/Week Limit:** \$50 sounds generous, but unoptimized prompting with premium models (Claude 3.5 Sonnet / GPT-4o) can deplete it in 3 days. Four team members debugging will quickly burn hundreds of API calls.

QUALITY RISK **Free-Form Grading for Chinese:** HSK Chinese is unforgiving with Pinyin, tones, Hanzi characters, and grammatical particles (了, 着, 过, 把-structures). Smaller models frequently fail at nuanced grading without rigid rubric enforcement.

3. Optimal Tech Stack for the 4-Week Event

Layer	Recommended Tech	Rationale & Practical Tip
Orchestration	LangGraph or pure Pydantic State	LangGraph StateGraph fits the proposal perfectly. If developers struggle with abstractions: a simple Python class model backed by SQLite persistence is completely sufficient!
Backend API	FastAPI + Pydantic v2	Pairs well with systems programming skills. Pydantic v2 guarantees strict schema validation for all LLM outputs.
Vector DB & Storage	ChromaDB + SQLite	ChromaDB runs embedded without extra Docker containers. SQLite stores user state, retention data, and session history reliably.
Frontend UI	Streamlit or Vite + Tailwind	For 4 weeks: use Streamlit for internal prototypes in Weeks 1–2. Build a custom React/Vue UI only if bandwidth permits.
Cloud LLM	OpenRouter API	Use DeepSeek-V3 / Qwen-2.5-72B for Chinese grading (extremely low cost & outstanding Chinese language capabilities) instead of costly GPT-4o.
Local Backup	Ollama with Gemma / Qwen2.5	Connect Gemma locally via Ollama/vLLM. Switch seamlessly via a unified OpenAI-compatible client interface (Base URL switch).

4. Strategic 4-Week Development Roadmap

Week 1: Data Models, SQLite & State Machine (No Frontend!)

- Define Pydantic schemas for LearnerState, ConceptMastery, and Exercise.
- Implement deterministic routing logic and mathematical retention decay algorithms.
- Set up OpenRouter and Ollama clients with toggleable base URLs.

Week 2: Grading & Teaching Loop (Core Agent Engine)

- Prompt engineering for rubric-based HSK grading returning structured JSON outputs.
- Populate ChromaDB with 30 curated HSK3–HSK4 grammar and vocabulary modules.
- Benchmark accuracy against 25 human-annotated reference sentences.

Week 3: FastAPI Backend & Asynchronous Persistence

- Integrate LangGraph into FastAPI endpoints.
- Decouple synchronous user interactions (low latency) from asynchronous state recalculations.
- Deploy a barebones Streamlit dashboard for internal team testing.

Week 4: Frontend Polish, Fallback Hardening & Final Demo

- Complete web UI (daily study plan view + dynamic progress bar).
- Ensure 100% offline demonstration capability (emergency fallback to local Gemma/Qwen via Ollama).
- Craft the final demo narrative and pitch slides for the jury.

5. Leadership & Mentoring Guidelines for the Team

- **Clear Interface Contracts (Interfaces First):** Define all Pydantic schemas on Day 1. Once inputs and outputs are immutable, all 4 team members can build in parallel without blockers.

- **Avoid the "Agent Framework Rabbit Hole":** Do not get entangled in multi-agent framework complexities. An agent is fundamentally: *Prompt + Tools + Schema + State*.
- **Chinese-Specific Model Routing:** Use **Qwen-2.5-72B-Instruct** or **DeepSeek-V3** via OpenRouter. These models outperform Western models in Chinese language logic at a fraction of the \$50 budget.